

1: Policy Data Issues & Assumptions

Issue Description	Assumption / Action Taken
14 duplicate entries: 8 exact row duplicates; 6 with conflicting Sum Assured values	Checked conflicting records against claims data (no claim history found). Retained the first occurrence of the data for all duplicates.
Policies with missing Issue Dates or Issue Dates falling outside the study period (Jan 2000 – Dec 2025)	Excluded these policies, as they fall outside the relevant scope and would prevent accurate exposure calculations.
Inconsistent casing and labels	Mapped invalid Occupation Classes to NA, as they are not crucial for calculating incidence rates.
Unit mixing, negative numbers, and dummy placeholders (Biometrics)	Set implausible height/weight records to NA, as they do not impact the incidence rate calculation.
Currency symbols, commas, negative values, and dummy placeholders (Financials)	Imposed realistic boundaries for the Sum Assured. Values outside this range were treated as NA.
Missing critical fields: 8 missing DOB, 8 missing Issue Date, 2 missing Sum Assured	Excluded policies missing critical dates from exposure calculations. Retained missing non-critical underwriting fields as NA to avoid unnecessary sample loss.

2: Claims Data Issues & Assumptions

Issue Description	Assumption / Action Taken

6 duplicate pairs	Retained the first occurrence for each duplicate pair.
Claims diagnosed after the study period	Excluded these claims.
Claims with blank diagnosis dates	Excluded these claims, as missing dates prevent us from verifying claim validity.
Inverted plan codes; missing product names	Populated missing product names using their corresponding plan codes from the policy dataset.
Non-critical conditions approved (e.g., Common Cold, Fracture)	Excluded these claims, as the conditions do not align with Critical Illness policy definitions.
Claim Status showing 'Declined' or 'Pending'	Excluded these claims, as they are not relevant for calculating the final incidence rate.
Critical Illness Riders showing multiple claims on the same policy	Enforced maximum claim limits based on policy type. CI Riders were restricted to a single claim per policy, while aggregate claims for base policies were capped at 300% of the Sum Assured.

3: Cross-Dataset Inconsistencies

Issue Description	Assumption / Action Taken
Claims where the Policy Number could not be matched to a valid record in the Policy dataset	Excluded these claims.

Policyholder Gender or Date of Birth differs between the Policy and Claims datasets	Treated the Policy dataset as the single source of truth and realigned the claims demographics to match.
Date of Diagnosis is earlier than, or within 90 days of, the Policy Issue Date	Excluded these claims to enforce the standard 90-day waiting period.
Claims where the policyholder was 75 years or older at the time of diagnosis	Excluded these claims based on the policy's maximum age limit.
Claims where the Date of Diagnosis occurred after the 'Status Date' for policies marked as 'Lapsed'	Excluded these claims, as the diagnosis occurred when the policy was no longer active.
"Early-stage" claims were marked as 'Paid' under the LifeSecure Rider	Excluded these claims, as the LifeSecure Rider strictly covers Major CI only.
Recorded Claim Amount exceeds the Sum Assured	Adjusted overpaid claim amounts down to the maximum allowed contractual limit, accounting for the yearly boosted Sum Assured benefits.